MATPLOTLIB

What is Matplotlib?

Matplotlib is a data visualization library in Python used to create graphs, charts, and plots.

What is PYPLOT?

Pyplot is a module of Matplotlib used to create graphs and plots easily in Python.

Why Pyplot is used

- To create graphs quickly
- To display data visually
- To add titles, labels, and colors to graphs

Why we use Matplotlib

Matplotlib is used to create graphs and charts in Python to visualize and understand data easily.

MATPLOTLIB SETUP

Learn how to start using pyplot.

Install:

```
pip install matplotlib
```

Import:

```
import matplotlib.pyplot as plt
```

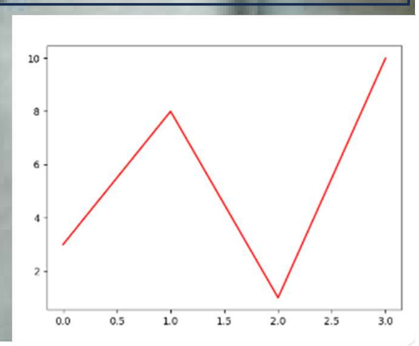
MATPLOTLIB GRAPH

1. **Line Plot**

   Show trends over time or continuous data.

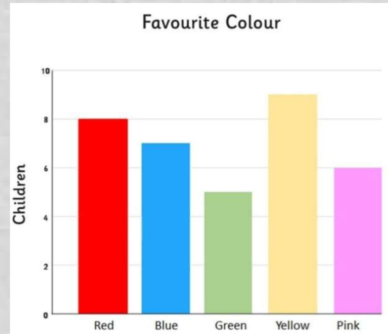
   Function:

   plt.plot()



2. Bar Chart

bar chart is a graph that uses rectangular bars to represent and compare different categories of data.



Function:

plt.bar()

Example

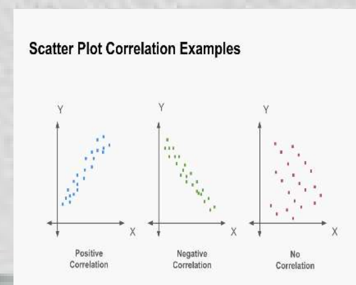
```
plt.bar(["A","B","C"],[10,20,15])
plt.show()
```

Variants:

- Vertical bar → `bar()`
- Horizontal bar → `barh()`

3. Scatter Plot

A **Scatter Plot** is a type of graph that shows the **relationship between two variables using dots**.

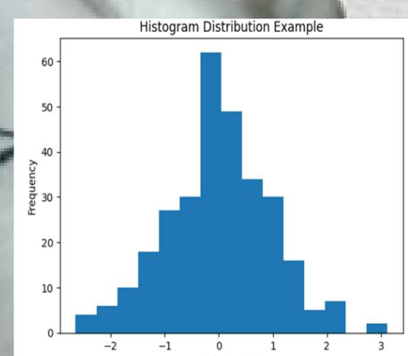


Function:

plt. scatter()

4. Histogram

A histogram is a graph that represents the frequency distribution of continuous data using adjacent bars



Function:

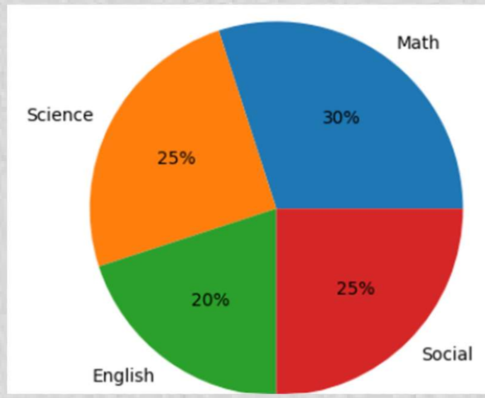
plt. hist()

5. Pie Chart

circular graph that represents data as **slices of a circle showing proportions or percentages**

Function:

plt. pie()

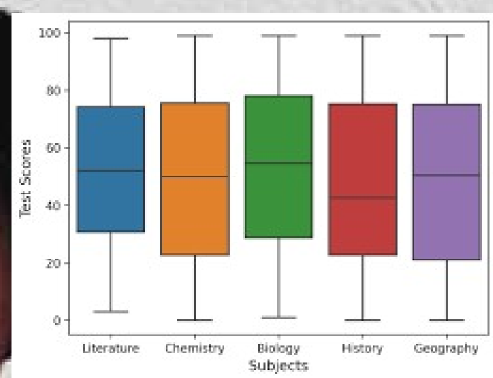


6. Box Plot

A Box Chart is a graph that displays the distribution of data using minimum, quartiles, median, and maximum values.

Function:

plt. boxplot()

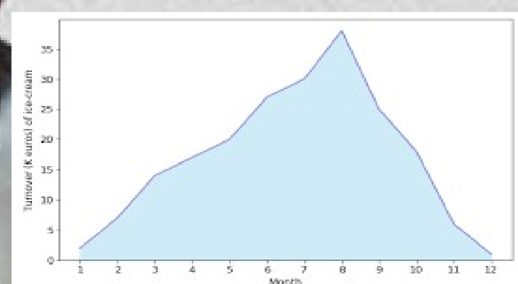


7. Area Plot

An Area Plot is a graph that displays data with the area below a line filled to show trends over time.

Function:

plt.fill_between(x, y)

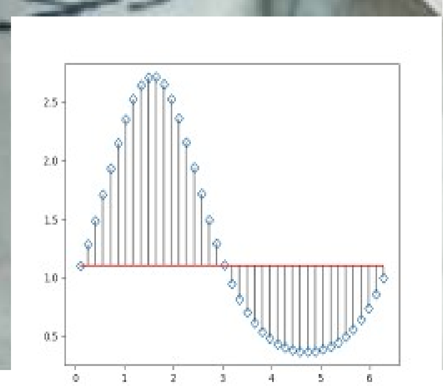


8. Stem Plot

A Stem Plot is a graph that displays discrete data values using vertical lines and markers.

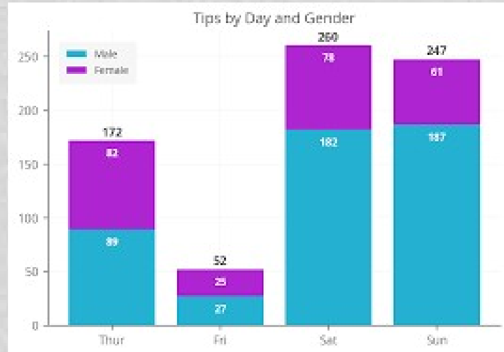
Function:

plt.stem(x, y)



9. Stack Plot

A stack plot is a graph that shows multiple datasets stacked on top of each other to represent the total and individual contributions



Function:

plt.stackplot(x, y1, y2, y3, ...)

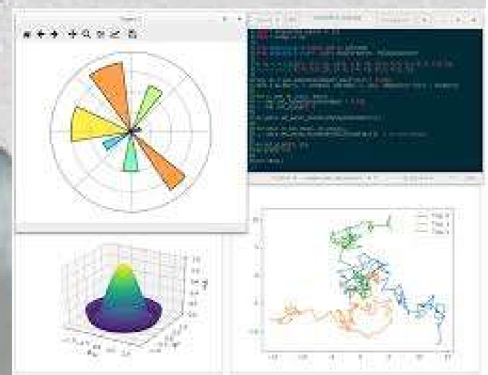
10. Heatmap / Image Plot

A heatmap is a graphical representation of data where values are represented using different colors

Function:

plt.imshow(data)

Image Plot: A graph used to display image or matrix data using the imshow() function in Matplotlib



Function:

plt.imshow(data)

MATPLOTLIB DISPLAY AND SAVE

What is Display Matplotlib?

Matplotlib display function used to show the plotted graph on the screen.

Function:

plt.show()

Example

```
x = [1,2,3,4]
y = [10,20,30,40]

plt.plot(x, y)
plt.show()
```

What is Display Matplotlib?

Matplotlib allows you to **save a graph as an image file** such as PNG, JPG, or PDF.

Function:

```
plt.savefig("filename.png")
```

Parameters in Matplotlib

Parameters are used to customize the graph such as color, labels, and line style.

1. **Color**

   Changes line color

2. **Label**

   Adds label for legend

3. **Linewidth**

   Changes line thickness

4. **Linestyle**

   Changes line style

5. **Marker**

   Adds markers on data points

Example with Parameters

```
plt.plot(x, y, color="red", linestyle="--", marker="o")
```

6. Dpi(Dots Per Inch)

It controls the **resolution (quality) of the saved image**.

- Higher DPI → clearer, higher quality image
- Lower DPI → smaller file, lower quality

Syntax

```python
plt.savefig("graph.png", dpi=300)
```

7. Transparent

The transparent parameter makes the background of the saved image transparent

Syntax

```
plt.savefig("graph.png", transparent=True)
```

8. Bbox(Bounding Box)

how much space around the figure is saved when saving a plot.

Syntax

```
plt.savefig("graph.png", bbox_inches="tight")
```

9. Format

Specifies the file format of the saved image.

Syntax

```
plt.savefig("graph", format="png")
```

10. Facecolor

Sets the background color of the figure.

Syntax

```
plt.savefig("graph.png", facecolor="white")
```

11. Edgecolor

Sets the edge color of the figure.

Syntax

```
plt.savefig("graph.png", edgecolor="black")
```

12. Orientation

Used to set the orientation of the saved image (mainly for PDF).

Syntax

```
plt.savefig("graph.pdf", orientation="landscape")
```

13. pad_inches

Adds extra padding around the saved figure.

Syntax

```
plt.savefig("graph.png", pad_inches=0.2)
```

PLOT IN MATPLOTLIB

WHAT IS PLOT?

Plot is a function used to visualize numerical data as a line graph.

Function:

plt.plot()

Syntax

```
plt.plot(x, y, color="red", linestyle="--")
```

Parameters

- X → Data for X-axis
- Y → Data for Y-axis
- Color → Line color
- Linestyle → Line style(- Solid line | -- Dashed line | : Dotted line | -. Dash-dot line)
- Linewidth → Thickness of line
- marker → Symbol for data points(o Circle | s Square | ^ Triangle | * | +
- label → Name shown in legend

SUBPLOT IN MATPLOTLIB

WHAT IS SUBPLOT?

Subplot means multiple graphs in one figure

Function:

plt.subplot()

Syntax

```
plt.subplot(rows, columns, index)
```

Parameters

- rows → Number of rows in figure
- columns → Number of columns
- index → Position of graph

Example

```
plt.subplot(2,2,1)
```

PLOT CUSTOMIZATION

WHAT IS TITLE?

Title is the heading of a graph that tells what the graph is about.

Syntax

```
plt.title("Title Name")
```

EXAMPLE

```
plt.title("Student Marks", fontsize=20, color="blue")
```

Parameters

- **Fontsize → change text size**
- **Color → change text color**
- **Loc → position of title**

WHAT IS AXIS LABELS?

Axis Labels are the names given to the X-axis and Y-axis in a graph

1. **X – AXIS LABEL**

   The X-axis is the horizontal line of the graph.

Syntax

```
plt.xlabel("Label Name")
```

2. **Y-Axis Label**

   The Y-axis is the vertical line of the graph.

Syntax

```
plt.ylabel("Label Name")
```

WHAT IS LEGEND?

A Legend is a box in a graph that explains what each line, color, or marker represents.

Syntax

```
plt.legend(handles=None, labels=None, loc='best', fontsize=None,
           ncol=1, frameon=True, shadow=False)
```

Parameters

Hendles → **Select specific lines to show in legend**

labels → **Custom legend names**

loc → **It is used to decide where the legend box will appear on the graph**

(best, upper right, upper left, lower right, lower left, center)

fontsize → **used to change the size of the text inside the legend.**

ncol → **ncol means number of columns in the legend**

WHAT IS GRID?

Grid is a set of **horizontal and vertical lines** on a graph

Syntax

```
plt.grid(visible=None, which='major', axis='both',
         color=None, linestyle=None, linewidth=None, alpha=None)
```

Parameters

Visible → **Show or hide the grid (True / False)**

Which → **which is used to decide which type of grid lines to show in the graph (major / minor/ both)**

axis → **used to choose which axis grid lines to display(both / x / y)**

color → **Grid line color (Red / blue / Green)**

linestyle → **used to change the style (pattern) of a line in a graph**

- **' - '** → **Solid line**
- **' -- '** → **Dashed line**
- **' : '** → **Dotted line**
- **' -. '** → **Dash-dot line**

alpha → **controls how transparent or visible a graph element.**

Axis and Ticks

What is Axis?

axis is a reference line in a graph.

1. **Xlim(X-axis)**

used to set or get the limits (range) of the X-axis

Syntax

```
plt.xlim(left, right)
```

Parameters

left → **Starting value of X-axis**

right → **Ending value of X-axis**

2. **Ylim(Y-axis)**

used to **set or get the limits (range) of the Y-axis**

- **Minimum value (start)**
- **Maximum value (end)**

Syntax

```
plt.ylim(bottom, top)
```

Parameters

bottom → **Minimum value of Y-axis**

Top → **Maximum value of Y-axis**

What are Ticks?

Ticks are the **small marks and numbers shown on the X-axis and Y-axis** of a graph

1. X-axis Ticks

used to set or customize X-axis values and labels.

Syntax

```
plt.xticks(ticks, labels=None, rotation=0)
```

Parameters

ticks → **List of position**

labels → **Custom labels for ticks**

Rotation → **Rotate labels (angle in degrees)**

2. Y-axis Ticks

controls the values and labels shown on the Y-axis.

Syntax

```
plt.yticks(ticks, labels=None)
```

Parameters

ticks → **List of Y-axis positions**
labels → **Custom names for those ticks**

Figure and Size Control

What are Figure?

The complete area (canvas) where the graph is drawn.

Syntax

```
plt.figure(figsize=(width, height))
```

Parameters

Figsize → **Size of figure (width, height in inches)**
dpi → **Resolution (dots per inch)**
facecolor → **Background color of figure**
edgecolor → **Border color**

What are Figsize?

controls the width and height of the graph.

Syntax

```
plt.figure(figsize=(width, height))
```

Parameters

Width → **Width of figure**
Height → **Height of figure**

Annotations

What are Annotations?

Annotation is used to add text and arrows to highlight specific points on a graph.

Syntax

```
plt.annotate(text, xy=(x, y), xytext=(x2, y2), arrowprops=dict())
```

Parameters

- text → Text to display
- xy → Actual data point
- xytext → Position of text
- arrowprops → Arrow style

What are Style?

Styles are ready-made design templates used to change the overall look of a graph

Syntax

```
plt.style.use("style_name")
```

Parameters

- ggplot → Grid + soft colors
- seaborn → Smooth and modern
- dark_background → Dark theme
- classic → Default style
- fast → Simple and fast rendering